Family-Leave Mandates and Female Labor at U.S. Firms: Evidence from a Trade Shock

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Abstract

We study the role of family-leave mandates in shaping the gender composition at U.S. firms that experience a negative demand shock. In a regression discontinuity framework, we compare firms mandated to provide job-protected leave under the Family and Medical Leave Act (FMLA) and firms that are exempt from the law (non-FMLA) following the post-2001 surge in Chinese imports. Using confidential microdata on matched employers and employees in the U.S. non-farm private sector, we find that between 2000 and 2003, an increase in import competition decreases the share of female workers at FMLA compared to non-FMLA firms. The negative differential effect is driven by female workers in prime childbearing years, with less than college education, and is strongest at firms with all male managers. We find similar patterns in changes in the female share of earnings and promotions. These results suggest that, when traditional gender norms prevail, adverse shocks may exacerbate gender inequalities in the presence of job-protected leave mandates.

Keyword*:* FMLA, job-protected leave, gender gap, regression discontinuity, China shock, gender norms

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1 Introduction

Family-leave programs offer workers the ability to balance career and family responsibilities. On the one hand, leave programs can be especially beneficial for women who tend to shoulder a larger share of family and child-care responsibilities.[1] On the other hand, leave programs may impose additional costs on firms. However, empirical evidence on the labor market impacts of family-leave programs has not yielded a clear consensus (Olivetti and Petrongolo, 2017). We make progress by providing causal evidence on the role of job-protected leave mandates in shaping the gender composition at firms through the lens of firm adjustments to a trade-induced negative demand shock.

We study the 1993 Family and Medical Leave Act (FMLA), a federal mandate which requires U.S. employers with 50 or more workers to provide twelve weeks of unpaid, job-protected family leave to qualifying employees. The firm size threshold provides a sharp regression discontinuity (RD) setting to study the impact of FMLA. Our focus is on understanding the role of FMLA in shaping U.S. firms' short-run adjustments to demand shocks. Therefore, we compare the gender composition at firms mandated to provide leave under FMLA with firms that are exempt from the law (non-FMLA), across sectors that were differentially exposed to the surge in Chinese imports following China's accession to the World Trade Organization (WTO) in 2001.

Our empirical analysis utilizes confidential, matched data on employers and employees in 2000 and 2003 for the U.S. non-farm, private sector sourced from the U.S. Census Bureau's Longitudinal Employer-Household Dynamics (LEHD). The data contain information on workers' quarterly earnings by employer and a set of demographic characteristics. We study changes in the share of female employment at the firm-state level combined with measures of changes in Chinese import penetration at the firm's six-digit North American Industrial Classification System (NAICS). We examine the female share of earnings and fe-

[1] See survey evidence in report by Pew Research Center (2013).

male share of promotions in subsequent analyses. Our main analysis sample, where firm FMLA status is determined solely by the employment criterion, includes firms that employ between 45 and 55 workers. We include all single-establishment firms. For firms with multiple establishments, we only include units where there is a single establishment in a given state to ensure comparability with the single-unit firms.[2] We validate the RD design by ruling out manipulation of employment around the cut-off; confirming that no other relevant policies apply at the 50-worker threshold; and ensuring that our results are robust to a larger bandwidth.

We find that a one standard deviation increase in Chinese import penetration lowers the share of female employment by about 5% at FMLA relative to non-FMLA firms. This relative decrease is strongest among female workers between the ages of 25 and 35, where fertility is highest and workers are most likely to have young children. Further exploration separating firms with female and male managers reveals a pronounced negative FMLA effect at firms with male compared to female managers. We also examine changes in the share of female earnings and promotions. We find that an increase in Chinese import penetration lowers the share of earnings accrued to female workers between 25 and 35 years at FMLA compared to non-FMLA firms. Conditional on firms promoting at least one employee, an increase in Chinese import penetration is also associated with a decrease in the female share of promotions at FMLA compared to non-FMLA firms.[3] The negative effects are statistically significant among female workers in prime childbearing ages and workers with less than college education.

We establish the robustness of our results using an alternate sharp regression discontinuity design that exploits the employee eligibility criteria under FMLA. Employees are eligible for leave if the employer employs 50 or more workers within a 75 mile radius of their work

[2]Restricting attention to a single establishment of a multi-unit firm in a given state permits identification of the FMLA status at the firm-state level.

[3]Promotion is defined as a 7% increase in annual earnings.

locations. We compare the gender composition at similarly sized firms that differ in their FMLA status only due to the distance rule. Results under the distance RD strategy are qualitatively similar to results under the employment RD strategy.

Our results are consistent with a framework where leave mandates impose costs on employers, associated with hiring temporary replacement workers or coordinating work schedules of existing employees. Family-leave mandates also facilitate the continuity of employment, thus conferring benefits in the form of higher retention of workers embodying firm-specific human capital, who may have quit in the absence of mandated leave. The surge in Chinese imports increases competition in product markets and firms experiencing a negative demand shock reduce overall output and labor demand (Autor, Dorn and Hanson, 2013). Smaller firms may find it more costly to accommodate workers on leave by either spreading tasks among existing workers or hiring replacement workers. If managers believe that women are more likely to avail of leave to fulfill care-giving responsibilities, the increase in expected leave provision costs due to greater import competition is magnified for women, resulting in a decrease in female relative to male employment.

We contribute to three strands of literature. First, we complement the nascent literature examining firm responses to family-friendly workplace policies. Rossin-Slater (2018) documents that there are few studies focusing on firm responses to maternity and family-leave policies. The handful of papers that have begun to examine the impacts of leave taking on U.S. firms have found no significant impacts on firms' total wage costs or turnover rates (Bedard and Rossin-Slater, 2016; Bartel, Rossin-Slater, Ruhm and Waldfogel, 2016). Minimal impacts of leave take-up on firms' total wage costs and employee turnover rates as well as output, gross profits and survival are corroborated in Danish data (Brenøe, Canaan, Harmon and Royer, 2020). However, these studies do not separately examine the impact of leave provision on relative employment, earnings, and promotions of female versus male workers.

Moreover, we provide one of the first set of results on the causal impact of job-protected family leave mandates on the firms' gender composition as they adjust to negative demand

shocks. The surge in Chinese import competition is an exogenous negative shock to demand because it was predominantly a result of increased productivity of Chinese firms and falling trade barriers post-WTO accession rather than changes in U.S. demand. The U.S. economy is increasingly global, and disruptions from trade wars, financial crises, natural disasters, and pandemics are pervasive. Thus, it is important to understand the firm level impacts of domestic family-leave policies during periods of adjustments.

Second, we contribute to the literature examining the impact of FMLA on women's labor market outcomes. Prior to the enactment of FMLA, 12 states mandated job-protected maternity leave. The FMLA is the first federal job-protected leave mandate requiring covered U.S. businesses to offer unpaid leave to qualifying workers.[4] It increased leave-taking by mothers and the likelihood of returning to work after childbirth (Baum, 2003a; Berger and Waldfogel, 2004; Han, Ruhm and Waldfogel, 2009; Klerman and Leibowitz, 1997). The policy has been associated with small increases or no effect on employment and small decreases in wages (Baum, 2003b; Ruhm, 1997; Klerman and Leibowitz, 1997; Waldfogel, 1999) but negative impacts on promotions (Thomas, 2019). Earlier work relied on self-reported survey data on wages and employer size and suffered from small sample sizes. We utilize administrative data on U.S. non-farm, private-sector employers and their workers, thereby providing a more comprehensive picture of the effects of FMLA. We emphasize that as firms adjust to negative shocks, leave mandates can shape the gender composition of employment, earnings and promotions.

Finally, our paper contributes to the literature on the impact of trade liberalization on gender gaps. Trade liberalization has generally reduced gender gaps in emerging economies while exacerbating them in developed countries. Gaddis and Pieters (2017) document that the comprehensive tariff cuts across Brazil led to a reduction in the gender gap through a

[4]Paid maternity leave in the U.S. is not directly job protected and can be traced back to state level short-term disability insurance programs (Timpe, 2018) and more recent paid family leave legislation such as in California (Bana, Bedard and Rossin-Slater, forthcoming; Bailey, Byker, Patel and Ramnath, 2019; Rossin-Slater, Ruhm and Waldfogel, 2013) and Rhode Island (Campbell, Chyn and Hastings, 2019).

larger relative decrease in male labor force participation and employment. Juhn, Ujhelyi and Villegas-Sanchez (2014) find that tariff reductions associated with the North American Free Trade Agreement (NAFTA) induced Mexican firms to enter export markets, adopt new technologies that lower the demand for physically demanding tasks, and replace blue-collar male with blue-collar female workers. Gender gaps in developed economies have tended to widen after trade liberalization. Hakobyan and McLaren (2018) find evidence of lower relative wage growth for married blue-collar women in areas of the U.S. that faced stiffest competition from Mexican imports following NAFTA. Autor, Dorn and Hanson (2015) find that areas in the U.S. that initially specialized in routine-intensive occupations experienced larger reductions in the employment-to-population rate of females compared to males in response to increased Chinese import competition. Keller and Utar (2018) find that an increase in Chinese import competition was associated with an increase in the gender gap in Denmark and attribute this to prime-aged women focusing more on children and family. Our results underscore the role of labor market policies in understanding differential outcomes for women exposed to trade shocks.

The rest of the paper is organized as follows. Section 2 describes the federal mandate and presents a conceptual model highlighting how a trade shock can shape female relative to male employment at covered firms. We describe our identification strategies and empirical specification in Section 3. Section 4 describes the confidential, matched employer-employee micro data used to construct the outcome variables of interest and the public-use data used to construct the measure of import penetration. We present results using the employment RD framework in Section 5. Section 6 discusses results from the distance RD framework. We explore additional outcomes in Section 7 and conclude in Section 8.

2 Policy Background and Conceptual Framework

2.1 Policy Background

The 1993 FMLA entitles eligible employees of covered employers to take up to twelve weeks of unpaid, job-protected leave in a 12-month period for family and medical reasons.[5] Covered employers refer to all private-sector firms with 50 or more workers.[6] An eligible employee must have worked for a covered employer for at least twelve months with a minimum of 1,250 hours of service preceding the leave. Moreover, the employee must be working in a location where the employer has 50 or more workers located within 75 miles.[7] About 47% of private-sector workers were both covered and eligible for FMLA leave in 2000 (Waldfogel, 2001).

Leave under FMLA can be taken for care of a newborn, adoption, foster care, care of a sick family member (spouse, child, or parent), or employee's own illness. The FMLA is intended to provide gender neutral access to leave such that both male and female workers can avail of leave under FMLA. Evidence suggests that female than male workers are more likely to utilize FMLA for care-giving purposes. In particular, women with young children are considerably more likely to be leave-takers relative to men with young children. In 2000, 76% (45%) of eligible women (men) with children 18 months or younger took leave under FMLA (Waldfogel, 2001). Moreover, the length of leave taken by women are more than twice as long as leave taken by men (Armenia and Gerstel, 2006).

A majority of covered firms report that they do not find compliance with administrative requirements of the mandate to be onerous and do not report negative impacts on their

[5]Military personnel may take up to twenty-six weeks of leave in a single 12-month period to care for a covered service member with a serious injury or illness. See 29 C.F.R. § 825 2019 for details on the law.

[6]Covered employers also include all public agencies and all public or private elementary or secondary schools. In this study, we focus on non-farm, private-sector employers.

[7]Employers are obligated to provide workers with general notice of FMLA rights. This includes displaying a general notice visible to all employees (Department of Labor, 2016).

productivity, profitability, or growth.[8] Nonetheless, firms do incur costs of providing leave to employees. The employer must continue providing any health insurance coverage to an employee on leave. The employer must also bear the cost of replacing the employee for the period of leave, either by hiring a temporary employee or by reallocating the employee's workload across other existing employees.

While leave provision under the FMLA may impose costs on firms, it can boost employee morale and increase employee retention. Over 80% of covered establishments, in 2000, reported that FMLA had a positive or no noticeable effect on employee productivity, absences, turnover, career advancement, and morale. In fact, more respondents reported that FMLA had a positive than a negative effect on employee morale: about 24% positive versus 11% negative (Department of Labor, 2000).

We present a conceptual framework in Section 2.2 incorporating these insights on costs and benefits of leave provision to trace out differential effects of the trade shock on firms covered by the FMLA compared to non-FMLA firms.

2.2 Conceptual Framework

We sketch a conceptual framework that analyzes the impact of import competition on relative employment of female workers in the presence of job-protected leave mandates. Consider a monopolistically competitive firm producing a differentiated product and operating as a price-taker in the labor market. We assume that the firm can employ workers at market wages for female and male workers of w_F and w_M, respectively. The production function exhibits constant elasticity of substitution with constant returns to scale. The firm minimizes total cost subject to its technology constraint as follows

[8]The Department of Labor's Survey of Establishments in 2000 documents that 64% of covered firms found it very or somewhat easy to comply with the administrative requirements of the FMLA. A vast majority of establishments reported that FMLA had no noticeable or a positive effect on their productivity (84%), profitability or growth (90%).

$$\min_{L_F, L_M} w_F L_F + w_M L_M + \lambda[(g\tau\theta(Y) - \gamma_F) w_F L_F + (g\theta(Y) - \gamma_M) w_M L_M] \tag{1}$$

subject to

$$Y \geq [(\alpha_F L_F)^\rho + (\alpha_M L_M)^\rho]^{\frac{1}{\rho}}. \tag{2}$$

Here, $0 < \rho < 1$ is the elasticity of substitution between male and female labor; α_F and α_M are female and male labor productivity, respectively; λ is equal to one for firms covered by the leave mandate (FMLA firms) and zero for exempt (non-FMLA) firms, for whom the expression in brackets that follows is irrelevant; g is the probability that a worker takes leave when she/he is called upon to fulfill care-giving responsibilities; $\tau > 1$ captures the firm's belief that female workers are more likely to take leave; and $\theta(Y)$ is the replacement cost of workers on leave. We posit that the replacement cost is higher for smaller firms, or $\frac{d\theta}{dY} < 0$. The idea is the following: if a worker is on leave, it is more costly to spread tasks across other workers when the firm operates at a smaller scale or hiring temporary replacement workers may represent a larger share of costs with reduced profits.[9] For instance, at larger scale, a firm may have more workers who can cover the work schedules of employees on leave.

The terms γ_F and γ_M refer to the firm's benefit from leave provision under the mandate. This benefit accrues to the firm even if the worker does not actually take leave. It may include gains from firm-specific human capital the firm retains by keeping the worker and greater worker effort because she/he values the option of taking leave to fulfill care-giving responsibilities when they arise. We allow the value workers place on the leave policy to differ by gender, with a prior that its value is larger for women.[10]

[9] Using a panel survey of California businesses in 2003 and 2008, Dube, Freeman and Reich (2010) estimate employee replacement costs relative to annual wages at between approximately USD 3,000 and 4,500 (in 2003 dollars) per recruit. On average, this represents about 10% of annual wage costs.

[10] This assumption is similar in spirit to Goldin, Kerr and Olivetti (2020) who examine why firms offer paid parental leave. In their model, the key source of gender differential is that women place a higher value on non-market time in expectation.

First order conditions of cost-minimization yield optimal relative female employment as

$$\frac{L_F^*(Y)}{L_M^*(Y)} = \left[\frac{\alpha_F}{\alpha_M}\right]^{\frac{\rho}{1-\rho}} \left[\frac{W_M(Y)}{W_F(Y)}\right]^{\frac{1}{1-\rho}} \tag{3}$$

where

$$W_F(Y) = w_F[1 + \lambda(\tau g\theta(Y) - \gamma_F)] \tag{4}$$

$$W_M(Y) = w_M[1 + \lambda(g\theta(Y) - \gamma_M)]. \tag{5}$$

Equation 3 shows that the relative employment of female workers is a positive function of their relative productivity (first right-hand side term) and a negative function of relative cost of employment (second right-hand side term). Total cost as a function of output is then

$$C(Y) = W_F(Y)L_F^*(Y) + W_M(Y)L_M^*(Y). \tag{6}$$

The firm's inverse demand is given by $P(Y)$. The firm then maximizes profit to set optimal output Y^*

$$\max_Y P(Y)Y - C(Y). \tag{7}$$

An increase in import competition impacts the relative employment of female workers through the second term, the relative cost of female workers, in Equation 3. Greater import competition lowers firm demand which lowers optimal output. The replacement cost of workers on leave is higher at lower levels of output. In addition, the expected cost is higher for women given the firm's belief, based on traditional gender norms, that women are more likely to take leave to fulfill care-giving responsibilities. As a result, an increase in import competition increases the relative cost of employing female workers in expectation and unambiguously decreases relative female employment at FMLA firms.[11]

[11] We provide a numerical example to illustrate this channel in Appendix A.

3 Empirical Framework

We now describe the sources of exogenous variation used to identify the impact of Chinese import competition on the gender composition at FMLA compared to non-FMLA firms and discuss our empirical specification.

3.1 Identification Strategy

Our identification strategy combines two sources of exogenous variation - a sharp RD in FMLA status and changes in Chinese imports across industries between 2000 and 2003. We employ two different sharp RD designs that separately exploit the employment and distance cut-off requirements to provide leave under FMLA.

For the employment RD design, we classify firms employing 50 or more workers as FMLA; and non-FMLA if they employ less than 50 workers. We address three main concerns about the employment-based RD strategy. First, we show that there is no evidence of firms manipulating employment to stay under the 50-worker threshold in the pre-shock period. Second, we confirm that there are no other federal policies that change sharply at the same cutoff. A comprehensive survey of business size thresholds in the application of federal statutes finds that no other federal policies apply at the 50-worker threshold (Keefe, Gates and Talley, 2005). Third, we address the concern that firms may change FMLA status between 2000 and 2003. Firms switching to non-FMLA status and vice versa subsequent to the increase in Chinese import competition would be a response to the shock. In fact, the inclusion of firms that switch FMLA status between 2000 and 2003 would result in an underestimation of our coefficient of interest, because we would attribute responses of non-FMLA firms to the FMLA group or vice versa. Further, results from the distance-based RD design, where switching is less of a concern, also provide robust support for our main finding.

For the distance-based RD design, we restrict our attention to firms employing 50 or more workers across two establishments (but fewer than 50 in each establishment), and classify

firms whose establishments are less than or 75 miles apart as FMLA, and establishments more than 75 miles apart as non-FMLA.

We fix a firm's FMLA status as of the pre-trade shock period, 2000, and compare FMLA to non-FMLA firms, before and after the surge in Chinese exports to the U.S. following China's accession to the WTO in 2001. This so-called "China" shock was predominantly a result of internal Chinese supply shocks and falling global trade barriers rather than U.S. demand shocks (Autor, Dorn and Hanson, 2013). Therefore, it is unlikely to have been correlated with characteristics of U.S. firms in our analysis samples.

3.2 Empirical Specification

We fit the following model:

$$\Delta Y_i = \beta_0 + \beta_1 FMLA_{i,2000} + \beta_2 \Delta IP_j + \beta_3 FMLA_{i,2000} * \Delta IP_j + \beta_4 f(r_i) + \beta_5 f(r_i) FMLA_{i,2000} + \beta_6 f(r_i) \Delta IP_j + \beta_7 f(r_i) FMLA_{i,2000} * \Delta IP_j + \beta_8 X_{i,2000} + \beta_9 X_{k,2000} + \delta_s + \Delta\varepsilon_i. \quad (8)$$

where the outcome variable of interest at firm i is denoted by Y_i. The firm is located in state s and operates in six-digit industry j within a four-digit industry k. Δ denotes change between 2000 and 2003.

The measure of Chinese import penetration within a six-digit NAICS industry j is given by IP_j. We also include a set of pre-shock controls. Firm i's FMLA status in 2000 is denoted by $FMLA_{i,2000}$ and picks up the effect at FMLA firms in industries that do not face Chinese import competition. $X_{i,2000}$ is a set of firm level controls as of 2000.[12] All regressions also include the share of female workers in a four-digit NAICS industry, $X_{k,2000}$, to control for the industry gender composition in the pre-shock period. We also include a set of state fixed effects, δ_s, to control for any state-specific policies that may also impact

[12]Under the employment RD framework, $X_{i,2000}$ includes firm age and multi-unit status. Under the distance RD framework, $X_{i,2000}$ includes firm age and employment.

the gender composition at firms. The rating variable, r_i, is centered around the cut-off such that under the employment cutoff, $r_i = (employment_i - 50)$ and under the distance cut-off, $r_i = (distance_i - 75)$. ε_i is an idiosyncratic error term and we cluster the standard errors at the four-digit NAICS level.

Our primary focus is on the female share of employment defined as follows:

$$\Delta Y_i = \left[\left(\frac{Y_{fc}}{Y_{(f+m)}} \right)_{i,2003} - \left(\frac{Y_{fc}}{Y_{(f+m)}} \right)_{i,2000} \right] * 100 \tag{9}$$

where Y_i denotes the share of female workers at firm i; Y_{fc} indicates the total number of female employees at firm i in category, c; and Y_{f+m} indicates the total number of employees, female and male, at firm i. We estimate the impacts separately by worker age and educational attainment categories. We group workers into mutually-exclusive age categories of ≤ 24, 25-35, 36-45, and 46+.[13] We further group workers into college (with bachelor's or higher degrees) and less than college (with less than bachelor's degree) educated.

Under the employment RD design, we focus on the sample of firms closest to the employment cutoff: firms with 45-55 workers. We choose this cutoff guided by the optimal bandwidth selection method in Imbens and Kalyanaraman (2012).[14] The procedure yields an optimal bandwidth choice of 3.5 and we implement a bandwidth of 5. Under the distance RD design, we focus on the sample of firms closest to the distance cut-off: firms with similar employment but where its establishments are 65-85 miles apart. The procedure yields an optimal bandwidth choice of 7.4 and we implement a bandwidth of 10. In both the samples, we choose a slightly larger bandwidth to allow for the maximum number of observations while still adhering closely to the suggested optimal.[15]

The control function, $f(r_i)$, is a continuous second-order polynomial function of the

[13] Age bins selected based on 1999 fertility rates in the United States (Ventura, Martin, Curtin, Menacker and Hamilton, 2001).

[14] We use Stata's rdob.ado program available at `https://imbens.people.stanford.edu/software`.

[15] Increasing the number of firms in the analysis also ensures compliance with Census Bureau disclosure avoidance rules.

rating variable on each side of the cut-off point. Our choice is guided by the AIC statistic, reported in Table 1, which is a goodness of fit measure and captures the bias-variance trade off in model choice (Lee and Lemieux, 2010). A higher-order polynomial is associated with lower bias, but larger variance given that more parameters are estimated. We provide the AIC values associated with the models using quadratic, cubic and quartic polynomials relative to the linear polynomial, for which the AIC value is zero. The model with the lowest AIC is preferred. Thus, a negative AIC value indicates better model performance over the linear polynomial. The two rows relate to the distance and employment RDs, respectively. For the distance RD, we find that the quadratic polynomial is preferred to the linear version, while for the employment RD the linear polynomial is preferred.[16] We employ a second-order polynomial across both the employment and distance RD specifications. A non-linear polynomial addresses the concern that adopting a larger bandwidth relative to the optimal may make our model susceptible to bias.

Under the employment RD, the control function captures the relationship between the outcome variables and firm size. Under the distance RD, the control function captures the relationship between the outcome variables and the distance between the firm's establishments. Including an interaction between the rating variable and FMLA status accounts for the fact that FMLA status may impact both the intercept and the slope of the regression line (Jacob and Zhu, 2012). We allow the slope to vary on the two sides of the cutoff.

The coefficient β_2 captures the impact of an increase in Chinese import competition on non-FMLA firms, while the coefficient β_3 isolates the differential effect of the trade shock on FMLA compared to non-FMLA firms. We expect that $\beta_3 < 0$, reflecting the hypothesis that greater import competition increases the replacement cost of workers on leave, and combined with firms' beliefs that women are more likely to use leave, results in a perceived increase in the cost of employing female workers.

[16]We also estimate our baseline employment RD specification using a linear polynomial and find that our results remain qualitatively unchanged as reported in Table 5

4 Data

This section describes the three main sources of data in our analysis. First, we use information on the gender composition of firms using linked employer-employee data. Second, we utilize information on the employment and location of individual establishments under the common ownership of a firm to determine FMLA status. Finally, we use publicly available information on output, exports, and imports at the six-digits NAICS level to construct the measure of Chinese import penetration.

4.1 Firms and Workers

We utilize the Longitudinal Employer Household Dynamics data for U.S. non-farm, private sector firms and their employees (McKinney and Vilhuber, 2014). Employers are identified at the state-level and for purposes of this study we use the term firm interchangeably with firm-state. Worker-level information includes quarterly earnings by employer and demographic characteristics including gender, age, and education. We restrict attention to workers with a strong labor force attachment and most likely to meet the FMLA employee eligibility requirements in both 2000 and 2003.[17]

We begin by restricting the sample to workers who have three or more quarters of reported earnings at an employer and then construct a measure of annual earnings for each worker. First, we compute average quarterly earnings using earnings in all quarters worked. Then, we only retain quarters where earnings are at least equal to half of the average quarterly earnings. We re-compute average quarterly earnings using only the quarters in which earnings meet this threshold. Finally, we annualize the adjusted quarterly earnings to construct annual earnings for a worker employed at a given firm. This restriction further ensures that lower earnings for a worker does not reflect periods of leave-taking. Once we create the

[17] We do not have leave-taking information for individual workers.

sample of workers, we construct firm-level aggregates of employment, earnings, and promotions by workers' gender, age, and educational attainment.

4.2 Employment-Based Firm FMLA status

We construct a FMLA status indicator for firms in 2000, prior to the trade shock. Firms are mandated to provide unpaid, job-protected leave to eligible employees under FMLA if they employ 50 or more workers. The employment threshold may vary by state. Some states mandate more generous employment thresholds and firms operating in these states must use the state-specific threshold to determine FMLA provision.[18] We exclude these states from our analysis to use a consistent size threshold for determining firm FMLA status.

Under statute 29 C.F.R. § 825.105 2019, a private employer is covered if it maintains 50 or more employees on the payroll during 20 or more calendar workweeks (which need not be consecutive) in either the current or the preceding calendar year. We use employment information from the Longitudinal Business Database (LBD) to construct a binary FMLA status indicator at the firm-state level. The LBD enables us to measure distance between establishments and their employment.

The LBD consists of data on all private, non-farm U.S. establishments in existence that have at least one paid employee (Jarmin and Miranda, 2002). The underlying source of the LBD is the Census Bureau's Business Register (BR) which is the most current and comprehensive database of U.S. establishments and companies in the United States (DeSalvo, Limehouse and Klimek, 2016). The number of employees in the BR is sourced from the employer's annual and quarterly federal tax returns.[19] The reported employment is an aggregated total of full- and part-time employees, salaried corporate officers and executives, and persons on paid absences (such as vacation or sick leave) for the pay period including

[18]The following states have a more generous employment threshold than the federal mandate: Maine (15+), Massachusetts (6+), Minnesota (21+), Oregon (25+), Vermont (10+), and Washington DC (20+).
[19]The IRS forms are 941, 943, and 944.

March 12 in a calendar year. This measure of employment most closely corresponds to the definition of employees who must be counted to determine employer coverage.[20]

Our main analysis sample, where FMLA status is based solely on the employment criteria, includes firms with a single establishment in a state to facilitate comparison of firms of similar sizes. A single-unit firm is assigned a non-zero FMLA status if it employs 50 or more workers and zero otherwise. A multi-unit firm is assigned FMLA status in a similar manner using its firm-state employment. Moreover, we ensure that the establishment is more than 75 miles away from all other establishments of the firm. Restricting attention to only firms with single establishments in a state ensures that the distance rule does not determine FMLA status. Using the distance rule could potentially result in FMLA firms where employment is less than 50 and invalidate the employment-based RD setup. The baseline analysis is conducted on a sample of firms that employ between 45 and 55 workers.

The employment-based FMLA indicator may be subject to measurement error. Firm employment is likely to fluctuate around the 50 worker cut-off over the course of a year that will not be reflected in our annual FMLA status indicator. Moreover, under the law, the applicable firm size is the one at the time when an employee claims leave benefits but which we cannot observe. Such misclassification of a FMLA firm as non-FMLA or vice-versa is likely to bias our estimate of the impact of a leave mandate towards zero. Our empirical methodology attributes the differential response of FMLA relative to non-FMLA firms to the family leave mandate. Misclassification will effectively mute the differential response, such that our coefficient estimate of β_3 would be an underestimate of the true impact of the leave mandate.

[20] 29 C.F.R. § 825.105 2019 states that any employee satisfying the following conditions must be counted: (i) work in the U.S.; (ii) name appears on payroll records, whether or not any compensation is received; (iii) on paid or unpaid leave; (iv) work at foreign firms operating in the U.S.; (v) part-time, temporary, seasonal, and full-time.

4.3 Distance-Based Firm FMLA status

In addition to considering an employment-based FMLA status, we exploit the 75 miles distance threshold specified under the employee eligibility criteria of the law to alternately define FMLA status based on distance. This definition is less prone to suffer from measurement error compared to a definition of FMLA status based on employment. A firm is mandated to offer leave under FMLA to eligible workers. An employee is only eligible if she works for an employer that employs more than 50 workers in a 75 mile radius. Therefore, workers at firms of similar sizes may either be eligible or not depending on the geographic distance between the business units of the employer.

We construct a firm FMLA indicator that is based solely on distance as follows. We only consider firms that have exactly two establishments each of which has fewer than 50 workers, but which added together exceed the 50-worker threshold. The focus on firms with two establishments offers the most straightforward way to determine firm FMLA status based on distance. The size restriction ensures that no single establishment is FMLA under the employment rule only, but that the firm overall meets the FMLA's size threshold. Then we compute the distance between the centroid of the establishments' zip codes.[21] We consider a firm to be subject to FMLA if its establishments are less than or 75 miles apart. Thus, for this sample, firm FMLA status is determined solely based on distance between the firm's establishments.

4.4 Chinese import penetration

We construct a measure of an industry's exposure to Chinese import competition as the change in the Chinese import penetration ratio for a U.S. manufacturing industry over the

[21] We use the NBER 2000 Zip Code Distance Database accessed at `https://data.nber.org/data/zip-code-distance-database.html`. Comprehensive information on establishments' geographic coordinates in 2000 is not available in the LBD.

period 2000-2003. The measure is defined as

$$\Delta IP_j = \frac{\Delta M_j^{US-China}}{Y_{j,1997} + M_{j,1997} - E_{j,1997}} * 100 \tag{10}$$

where $\Delta M_j^{US-China}$ is Chinese imports to the U.S. by six-digit NAICS industry j; $M_{j,1997}$, $E_{j,1997}$, and $Y_{j,1997}$ are total U.S. imports, total U.S. exports, and total U.S. domestic production in 1997 by six-digit NAICS industry j, respectively. Δ denotes change between 2000 and 2003.

We obtain information on imports and exports from Schott (2008) and domestic output from Becker, Gray and Marvakov (2016). This measure of exposure to trade competition was introduced in Bernard, Jensen and Schott (2006) and used to study the China shock in Acemoglu, Autor, Dorn, Hanson and Price (2016). Table 2 reports the mean and standard deviation, and the 25th, 50th, and 75th percentiles of the distribution of the growth in Chinese import penetration between 2000 and 2003 across all three-digit NAICS manufacturing sectors. The sectors that experienced the highest average growth are primarily labor-intensive sectors, such as leather, textile products, furniture, and toys, in which China tends to have a comparative advantage. Average growth in import penetration was lowest for sectors such as beverages and tobacco, petroleum, metals and transport equipment. For non-manufacturing sectors, the exposure to Chinese import penetration is zero.

4.5 Summary statistics

Table 3 provides descriptive statistics of our main analysis sample, firms that employ between 45 and 55 workers, separately, for FMLA and non-FMLA firms in 2000. We compare the following outcomes: total number of female workers as a percentage of total workers at the firm, total earnings of female workers as a percentage of total earnings of all workers at the firm, total number of female promotions as a percentage of total promotions at the firm, total number of female workers, total number of male workers, firm age, and revenue per worker.

The two columns display the mean values and associated standard errors in parentheses and the third column shows the difference in the means of the variables at FMLA and non-FMLA firms and t-statistics in parentheses.

From the first three rows, we find that in 2000, FMLA and non-FMLA firms are very similar in their share of female workers at roughly 40%. This also holds for the share of female earnings at roughly 34% and for the share of female promotions at 40%. These results are consistent with Waldfogel (1999). She finds no significant effect on female employment in states without prior laws providing rights to a job-protected maternity leave compared to those with prior laws. The similarity in outcomes can be rationalized if prior to the trade shock, the retention benefit associated with the leave policy outweighed the replacement cost of a worker on leave.

Rows four and five show that the average FMLA (non-FMLA) firm in our sample employ about 19 (16) women and 28 (24) men. The difference in total female and male workers at FMLA and non-FMLA firms is statistically significant. However, this is not surprising given that FMLA status is determined by firm size. From row six, we see that non-FMLA and FMLA firms tend to be similar in age. In row seven, we find that FMLA firms are more likely to be multi-unit firms. Finally, from row seven, we find no difference in revenue per worker (a measure of firm performance) between FMLA and non-FMLA firms.[22] In our baseline regressions, we control for firm age and multi-unit status as of 2000 and also confirm the robustness of our results using the sample of single-unit firms only.

5 Baseline identification: Employment RD

In this section, we present results from our main empirical strategy: employment RD. We first confirm that manipulation around the employment cut-off is not a serious issue in the

[22]Revenue per worker is calculated on the sample of single-unit firms only because the LBD does not contain revenue information at the establishment level.

context of our study by performing the McCrary density test (McCary, 2008). We find that the log difference in the height of the density function of employment around the cutoff is 0.072 with a standard error of 0.064. Thus, the t-test of the null hypothesis of continuity cannot be rejected at conventional levels of statistical significance.[23]

Table 4 presents the baseline results from estimating Equation 8 under the employment RD framework. All regressions include state fixed effects, share of female employment in a 4-digit NAICS, controls for firm age and the first three columns additionally control for firm multi-unit status. The RD trade interaction coefficient, $FMLA \times \Delta IP$, isolates the response of firms mandated to provide leave under FMLA relative to non-FMLA firms to a change in Chinese import competition. The coefficient on ΔIP captures the impact of import competition on the female share of employment at non-FMLA firms. The first column shows the results for all firms in the baseline sample. The second column includes manufacturing firms only; the third column includes single-unit firms only; and the final column includes only multi-unit firms with a single establishment in a state. In all the samples, we focus on firms employing between 45 and 55 workers.

We find that the RD trade interaction coefficient is negative across all columns. From the first column, a one percentage point increase in Chinese import penetration is associated with half a percentage point lower share of female employment at FMLA relative to non-FMLA firms. The standard deviation of ΔIP is 4.34 as reported in Table 2. Therefore, a one standard deviation increase in ΔIP is associated with over 2 percentage points decrease in the female share of employment at FMLA firms relative to non-FMLA firms. On a base of about 40% from Table 3, this is approximately a 5% decrease. There is also wide variation across sectors. In high exposure sectors, such as Leather, a one standard deviation change in ΔIP results in a 4.4 percentage points decrease in the female share of employment at FMLA

[23] Appendix Figure A.3 provides a count of the number of firms within each employment bin, from 45 to 55. We do not observe any significant bunching at size bins just below the cut-off. There are, in fact, more firms that employ 50 workers relative to bins right below the cut-off.

firms relative to non-FMLA firms. The negative differential impact remains statistically significant and similar in magnitude using the sample of manufacturing firms only (second column) and sample of single-unit firms only (third column). In the last column, the RD trade interaction coefficient for multi-unit firms only, although imprecisely estimated, is much larger and negative.

We find that at non-FMLA firms, greater import competition is associated with a higher share of female employment. While explaining this effect is beyond the scope of this paper, we offer two possibilities. First, increased import competition may increase the relative productivity of female workers. The trade literature has firmly established the relationship between trade liberalization and technology upgrading where firms respond to import competition by investing in superior technology and improving productivity (Bustos, 2011; Topalova and Khandelwal, 2011). Modern technology complements brain, where women have a comparative advantage, relative to brawn (Do, Raddatz and Levchenko, 2016; Rendall, 2013). Second, if non-FMLA firms also provide some unpaid leave, they can adjust leave provisions in response to greater import competition.[24] With lower leave provisions, female workers may no longer be more costly to employ, resulting in increased female share of employment at non-FMLA firms.

5.1 Validation of Specification

In this section, we test the robustness of our baseline specification in the first column of Table 4. We present the results in Table 5. In the first column, we check the robustness of the baseline specification using a linear control function in the rating variable. The second column shows that our results are robust to the choice of bandwidth. We include firms employing between 40-60 workers. Across both columns, we find that our baseline results remain qualitatively similar.

[24]By 2000, about one-third of U.S. businesses not covered by the FMLA offered unpaid leave (Waldfogel, 2001).

In the last two columns, we report results from a set of placebo regressions. We test whether there exists a differential response by FMLA and non-FMLA firms applying a pseudo employment cut-off to determine FMLA status. We examine a cut-off of 40 using a sample of single-unit firms employing between 35 and 45 workers where a firm is defined as FMLA if it employs 40 or more workers and non-FMLA otherwise. We also examine a cut-off of 60 using a sample of single-unit firms employing between 55 and 65 workers where a firm is defined as FMLA if it employs 60 or more workers and non-FMLA otherwise. The idea is that if our identification strategy correctly isolates the effect of FMLA, we should observe no effects in the two placebo regressions. Indeed, the coefficients of ΔIP and $FMLA \times \Delta IP$ are indistinguishable from zero.

5.2 Net employment growth

A key question is whether the relative decline in the female share of employment at FMLA firms in response to the trade shock is driven by a decrease in female employment or an increase in male employment. We explore the net employment growth at FMLA compared to non-FMLA firms following the trade shock and decompose it by male and female employment. We measure net employment growth as follows:

$$\Delta Y_i = \left[\frac{Y_{i,2003} - Y_{i,2000}}{0.5 * (Y_{i,2003} + Y_{i,2000})} \right] \tag{11}$$

where Y_i is employment at firm i. This growth rate measure is standard in analysis of business dynamics (Davis, Haltiwanger and Schuh, 1998). We report the results in Table 6. Focusing on the first column, we find that a one percentage point increase in Chinese import penetration decreases net employment growth rate by 0.02 at FMLA relative to non-FMLA firms. The next two columns decompose the net growth by female and male employment, respectively:

$$\Delta Y_{ic} = \left[\frac{Y_{ic,2003} - Y_{ic,2000}}{0.5 * (Y_{i,2003} + Y_{i,2000})} \right] \tag{12}$$

where c is female or male employment, and ΔY_{ic} is the net female (or male) employment growth. Even though both male and female employment growth decline between 2000 and 2003, the decline in employment growth at FMLA compared to non-FMLA firms is driven almost exclusively by the decline in female employment growth.

5.3 Mechanisms

Our baseline results indicate that FMLA firms exhibit lower shares of female employment in response to a trade-induced negative demand shock. In this section, we probe the mechanisms underlying this response.

5.3.1 The role of worker age

Since a worker's likelihood of fulfilling care-giving responsibilities, particularly childcare, is higher for women in prime childbearing years, we expect the negative effects to vary systematically by worker age. We analyze the share of female workers decomposed into the following age categories: 24 and below, 25-35, 36-45, and 46 and older. Women between ages 25-35 are in their prime, childbearing years and are also more likely to have young children. In 2000, the average age of mothers at first birth is about 25 and at last birth is about 32 (Matthews and Hamilton, 2016). Older women than men are also disproportionately engaged in eldercare and hence may require leave (He, Weingartner and Sayer, 2018).

Table 7 shows that the negative employment effect at FMLA firms is strongest among female workers in the 25-35 age group and to a lesser extent in the 36-45 age group. These results are consistent with the idea that the gender differential between the perceived likelihoods of taking leave to fulfill care-giving responsibilities is higher for female workers in their

prime childbearing ages. We do not find any economically significant impacts for women aged 46 and older.

5.3.2 The role of manager's gender

Beliefs about gender roles can shape women's labor market outcomes (Charles, Guryan and Pan, 2018). We extend our analysis by asking if responses to greater import competition differ across firms with and without women in decision-making roles. If women in decision-making roles are less disposed to beliefs founded on traditional gender norms and more likely to enact family-friendly workplace policies, the impact of the trade shock on changes in the relative leave costs may be mitigated at firms with female managers.

We examine this hypothesis by identifying firms that have female managers. A firm is defined as female-managed if at least one of the top three earners at the firm is female. The underlying idea is that workers in decision-making roles are more likely to be highly paid. Prior studies using LEHD have identified firm owners including owners who actively manage the business using information on top three earners (Babina and Howell, 2018; Azoulay, Jones, Kim and Miranda, 2020; Kerr and Kerr, 2009). Given our focus on firms employing between 45 and 55 workers, this definition is more likely to identify the decision-makers than applying this definition to large firms that may have multiple layers of management. The results are presented in Table 8.

In the first two columns of Table 8, we find that the share of female employment is negative and statistically significant at FMLA compared to non-FMLA firms with all male managers while the effects are statistically insignificant at firms where at least one manager is female. In the last two columns, we redo the analyses for manufacturing firms only. By focusing on the manufacturing sector, we remove any heterogeneity in the proportion of female-managed firms across manufacturing and service sectors should they exist. We can see from the table that about 48% of the firms in the manufacturing sample and 45% in the full sample have at least one female manager. The negative differential effect for FMLA

firms persists at male-managed manufacturing firms. These results suggest that women in positions of power might facilitate more family-friendly work spaces and underscore the rationale for gender quotas on corporate boards (Bertrand, Black, Jensen and Lleras-Muney, 2019).

6 Alternate identification: Distance RD

We employ an alternate identification strategy exploiting the 75 miles distance threshold specified under the employee eligibility criteria of the law to define FMLA status based on distance. Under the distance RD, we compare the gender composition at similarly sized firms mandated to provide leave under FMLA with firms that are exempt from the law (non-FMLA), across sectors that were differentially exposed to the surge in Chinese imports following China's accession to the WTO in 2001.

We present results in Table 9. We find that, in the first column, the distance RD trade interaction coefficient is negative and statistically significant, though orders of magnitudes larger compared to the employment RD trade interaction coefficient in column one of Table 7. A one percentage point increase in Chinese import penetration is associated with a 10 percentage points decrease in the share of female employment at FMLA relative to non-FMLA firms. The difference in the magnitudes is likely driven by the distinct samples used: single-unit firms and multi-unit firms with a single establishment in a given state under the employment RD; multi-unit firms with exactly 2 establishments under the distance RD. The $FMLA \times \Delta IP$ coefficient under the employment RD, -3.782, for the multi-unit firms sample in Table 4 is more comparable.

The next four columns decompose the share of female employment by four mutually exclusive age groups. We find that the negative FMLA effect is statistically significant for workers in the 36-45 age group while the largest effect is for workers aged 24 or less. Overall,

our main finding - FMLA firms exhibit lower shares of female employment after the trade shock - remains qualitatively robust under this alternate identification strategy.

7 Extensions

We have, thus far, examined changes in the female share of employment at FMLA compared to non-FMLA firms across sectors facing varying degrees of Chinese import competition. Here, we present results on the female share of earnings and promotions and explore the heterogeneity in the impacts of leave mandates by worker's education levels.

7.1 Female share of earnings and promotions

We present results for the female share of earnings and promotions differentiated by age groups in Tables 10 and 11, respectively.[25] The LEHD does not contain information on hours of work. Therefore, we cannot separate out the wage and hours channels. In Table 10 we find the differential impact of import competition on the female share of earnings at FMLA firms is negative and statistically significant for female workers in the 25-35 age group.[26] A one percentage point increase in import penetration is associated with a 0.17 percentage point decrease in the share of earnings of female workers aged 25-35 years at FMLA firms.[27] However, due to data constraints, we cannot identify if the overall reduction in the female share of earnings is being driven by reduced hours or decrease in wages.

Table 11 presents results for baseline Equation 8 where the outcome variable is the share of female promotions in total promotions, conditional on firms promoting at least one employee, separately by age bins. A worker is defined as being promoted if her annual

[25]Results for share of female promotions and earnings are robust to using the sample of manufacturing firms only and single-unit firms only. These regressions are available upon request.

[26]Gruber (1994) found that mandated coverage for childbirth in health insurance policies was associated with lower wages for married women of childbearing age.

[27]The total FMLA effect is calculated as (-0.288 + 0.114).

earnings increase by at least 7%.[28] From the first column, we find that a one percentage point increase in Chinese import penetration is associated with a 1.6 percentage points decrease in the female share of promotions at FMLA relative to non-FMLA firms. This negative impact of the leave mandate in response to the trade shock is concentrated among female workers in prime childbearing ages of 25-35. These results are consistent with Thomas (2019), who finds that women hired after FMLA was enacted are less likely to be promoted than those hired before FMLA and the effect is driven by women 40 years or younger. In a framework where employers must choose to invest in worker training, uncertainty on future work hours of female employees after childbirth results in a gender wage gap that is exacerbated by mandated family leave.

7.2 The role of education

Import competition may have different impacts on workers across the skill distribution. The benefit of retaining higher-skilled workers would be greater as they are more likely to accrue firm-specific human capital, for instance, by receiving on-the-job training (Altonji and Spletzer, 1991). If the retention benefit outweighs the replacement cost of a worker on leave, we would expect a more muted impact for higher-skilled workers.

We proxy for skill using information on worker's educational attainment. We can observe a worker's level of educational attainment as of age 25 from the LEHD.[29] A shortcoming of the education data is that for the vast majority of workers, approximately 92%, it is imputed (see Abowd, Stepehens, Vilhuber, Andersson, McKinney, Roemer and Woodcock (2009) for details on the imputation method). Therefore, we consider the results by educational attainment as mostly being suggestive.

[28] McCue (1996) discusses studies that find that wage growth is 6 percentage points higher after a promotion and the average gain in wages is about 8%.

[29] LEHD does not contain information on worker's occupations.

We separate workers into two groups: with bachelor's degree or higher (high-skilled) and with less than bachelor's degree (low-skilled) and estimate our baseline specification by age group. We exclude results for workers aged 24 or younger, since educational attainment is measured at age 25. Panels A and B of Table 12 show results for college- and less than college-educated workers, respectively. Panel B shows a negative effect of Chinese import penetration on the share of female employment in FMLA compared to non-FMLA firms in the 25-35 age group. However, for workers with college or more education, shown in Panel A, we find no significant differential effects. In Appendix Table A-1, we show that these results are qualitatively robust to using the distance RD estimation strategy.

In Table 13, we decompose results for the female earnings share into results for college- (Panel A) and less than college-educated (Panel B) workers. The negative effect of greater import competition on the relative share of earnings for female workers is concentrated among less than college-educated workers in the 25-35 age group in Panel B. This is remarkably consistent with the results for employment. The impacts of import competition on relative outcomes for women are driven by women with less than college education and women in their childbearing years. The impact of leave mandates are likely to be mitigated for college-educated workers, whose firm-specific human capital might make retaining them more beneficial to the firm.

In Table 14, we decompose the share of female promotions by age and skill group. Again, we exclude individuals below 25 years of age. Results in bottom panel of Table 14 show that the impact of Chinese import penetration on the share of promotions of female workers with less than college education at FMLA firms is large, negative, and statistically significant in the 25-35 age group, where female fertility is at its highest. However, we find no statistically significant differential impacts of import penetration on FMLA firms in Panel A of Table 14, which focuses on college-educated workers.

8 Conclusion

This paper estimates the causal, short-run differential impact of a trade-induced negative demand shock on the gender composition at firms mandated to provide job-protected family leave compared to firms exempt from the law. Our results highlight the role played by the 1993 Family and Medical Leave Act in shaping relative outcomes for female workers as firms adjust to exogenous economic shocks in product and labor markets. Our study underscores the importance of considering the firm level impacts of domestic family-leave policies in an economy where pervasive shocks necessitate frequent adjustments at firms.

We find that FMLA firms relatively more exposed to the surge in Chinese import competition have lower female shares of employment, earnings, and promotions compared to non-FMLA firms. The negative differential effect is observed for female workers in prime childbearing years and with less than college education, for whom firm-specific human capital might be less relevant. Moreover, the negative effects at FMLA firms are stronger when all the managers are male.

We present a conceptual framework where employers trade off the replacement cost of workers taking leave with the benefit of retaining firm-specific human capital of the worker who might otherwise have quit to fulfill care-giving responsibilities. Increased import competition acts as a negative shock, inducing firms to contract overall. Smaller scale makes it costlier to replace workers, for instance, by temporarily assigning other workers to cover the tasks of the worker on leave. If managers believe that women are more likely to take leave, then the relative perceived cost of leave-taking by women will increase with greater import competition.

Our findings suggest that while there was no difference in the gender composition of employment between FMLA and non-FMLA firms before the trade shock, an increase in import competition from China is associated with a significantly lower share of female employment in FMLA relative to non-FMLA firms. Within our framework, the perceived replacement costs for female relative to male employees on leave, stemming from differential beliefs in the

likelihood of taking leave between female and male employees, increases due to the shock. The implication is that adverse economic conditions may generate gender inequalities, even with gender-neutral leave policies, in the presence of traditional gender norms. Leave policies may achieve greater gender neutrality, in practice, with complementary action such as increasing women's representation in decision-making roles to challenge existing attitudes towards gender. Indeed, if traditional gender roles shape managers' responses to negative economic shocks in the presence of family-leave mandates, our findings are also salient for paid leave mandates that are becoming more prevalent across different U.S. states.

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Table 1. Specification tests

	Quadratic	Cubic	Quartic
Share of female employment (Distance RD)	-4.8	2.7	8.6
Share of female employment (Employment RD)	6.1	10.4	15.3

Notes: This table reports the Akaike Information Criterion (AIC) from estimating Equation 8 using linear, quadratic, cubic, and quartic polynomials for the control function, $f(r_i)$, in the rating variable. The AIC is a goodness of fit measure and captures the bias-variance trade off in model choice. The three columns present AIC values for each n-order polynomial relative to the linear. AIC for the linear model is coded as 0, therefore, a positive (negative) AIC value indicates preference for a linear (n-order) polynomial.
Source: Authors' calculations using LEHD and LBD.

Table 2. Summary Statistics: Growth in Chinese Import Penetration (%)

Manufacturing Sector	mean	sd	p25	p50	p75
Food	0.243	0.836	0.001	0.024	0.113
Beverage and Tobacco	0.004	0.013	-0.002	0.002	0.008
Textile Mills	0.941	1.436	0.035	0.160	1.594
Textile Product Mills	4.202	3.673	1.093	3.763	6.540
Apparel	4.070	7.631	0.433	1.706	6.296
Leather	6.366	8.259	2.833	4.673	9.833
Wood	0.824	1.104	0.019	0.138	1.419
Paper	2.161	6.519	0.045	0.089	0.350
Printing	2.410	2.558	0.000	2.667	3.261
Petroleum and Coal	0.007	0.006	0.000	0.010	0.011
Chemical	0.367	0.482	0.039	0.191	0.562
Plastics and Rubber	0.758	0.967	0.114	0.207	1.132
Nonmetallic Mineral Product	0.748	1.402	0.005	0.131	0.857
Primary Metal	0.113	0.730	-0.211	0.008	0.274
Fabricated Metal	1.311	1.732	0.069	0.802	1.891
Machinery	1.782	5.988	0.150	0.304	1.154
Computer and Electronic Product	2.040	4.956	0.196	0.711	2.059
Electric Equipment	2.787	4.347	0.218	0.673	2.595
Transport Equipment	0.036	1.487	0.004	0.074	0.174
Furniture	4.727	5.816	0.323	2.683	6.098
Toys and Miscellaneous	3.658	6.486	0.386	2.799	5.574
All	1.645	4.340	0.025	0.237	1.444

Notes: Columns show the mean, standard deviation and the 25th, 50th and 75th percentiles of the growth in Chinese import penetration between 2000 and 2003 across 3-digit NAICS manufacturing sectors. Chinese import penetration is defined as in Equation 10.
Source: Author's calculations using public-use imports and exports from Schott (2008) and public-use domestic output from Becker, Gray and Marvakov (2016).

Table 3. Summary Statistics, Baseline Sample, 2000

	FMLA	non-FMLA	Difference
Share of female employment	40.62	40.70	0.0738
	(26.10)	(26.25)	(0.25)
Share of female earnings	34.14	34.13	-0.001
	(25.41)	(25.47)	(0.00)
Share of female promotions	40.57	40.49	-0.792
	(29.08)	(29.52)	(0.22)
Female workers	19.72	16.45	-3.277***
	(113.3)	(12.25)	(3.55)
Male workers	28.73	24.59	-4.139***
	(116.20)	(13.90)	(4.37)
Firm age	14.73	13.98	-0.758***
	(8.434)	(8.429)	(7.84)
Multi-unit status	0.0858	0.0135	-0.723***
	(0.280)	(0.115)	(29.45)
Firm revenue per worker[a]	167.10	167.20	1.63
	(255.40)	(264.00)	(0.48)

Notes: The first two columns display the means with standard deviations in parentheses for a select set of variables at FMLA and non-FMLA firms, respectively. FMLA status is based on the firm size threshold. The last column shows the difference in the variables at FMLA and non-FMLA firms with t-statistics in parentheses.

[a] Revenue per worker is calculated for the sample of single-unit firms only. Revenue information is only available at the firm level and, therefore, cannot be allocated across establishments for the sample of multi-unit firms.

Source: Authors' calculations using LEHD and LBD.

Table 4. Change in Female Share of Employment
Employment RD

	All	Manufacturing	Single-Unit	Multi-Unit
$FMLA$	-0.000	0.649	-0.022	4.380
	(0.409)	(0.831)	(0.410)	(2.851)
ΔIP_j	0.368*	0.493**	0.329*	4.159
	(0.190)	(0.187)	(0.192)	(3.287)
$FMLA \times \Delta IP_j$	-0.532**	-0.545**	-0.566**	-3.782
	(0.250)	(0.256)	(0.230)	(3.329)
Observations[a]	30,500	5,500	29,000	1,500

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of firms that employ 45-55 workers. The first column presents results on the full sample, and separately for manufacturing only, single-unit only, and multi-unit only firms. The dependent variable is defined as in Equation 9. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

Table 5. Change in Female Share of Employment, Specification Checks
Employment RD

			Placebo Cutoffs	
	Linear	40-60	40	60
$FMLA$	-0.007	-0.004	1.595	-2.595
	(0.240)	(0.282)	(2.721)	(2.728)
ΔIP_j	0.103	0.136	-0.058	-0.236
	(0.116)	(0.156)	(1.556)	(1.460)
$FMLA \times \Delta IP_j$	-0.231**	-0.269*	-0.008	-1.670
	(0.117)	(0.161)	(1.490)	(1.460)
Observations[a]	30,500	60,500	48,500	19,000

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. The first column displays results from estimating Equation 8 using a linear control function $f(r_i)$ in the rating variable on the sample of firms that employ 45-55 workers. The second, third, and fourth columns show results from estimating Equation 8 using a quadratic control function $f(r_i)$ in the rating variable on the sample of firms that employ 40-60, 35-45, and 55-65 workers, respectively. The dependent variable is defined as in Equation 9. All regressions include pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

Table 6. Change in Net Employment Growth
Employment RD

	All	Male	Female
$FMLA$	0.018	0.010	0.009
	(0.018)	(0.013)	(0.008)
ΔIP_j	0.005	-0.004	0.008*
	(0.011)	(0.008)	(0.005)
$FMLA \times \Delta IP_j$	-0.021**	-0.002	-0.019***
	(0.010)	(0.008)	(0.004)
Observations[a]	30,500	30,500	30,500

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of firms that employ 45-55 workers. The first column shows the total change employment as defined in Equation 11. The next two columns decompose the change by male and female employment, as defined in Equation 12, respectively. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

Table 7. Change in Female Share of Employment, Role of Age
Employment RD

	All	24 or less	25-35	36-45	46+
$FMLA$	-0.000	-0.099	-0.354	0.529**	-0.077
	(0.409)	(0.225)	(0.305)	(0.265)	(0.308)
ΔIP_j	0.368*	-0.114	0.226*	0.289*	-0.033
	(0.190)	(0.107)	(0.122)	(0.167)	(0.239)
$FMLA \times \Delta IP_j$	-0.532**	-0.050	-0.374***	-0.306*	0.197
	(0.250)	(0.097)	(0.118)	(0.176)	(0.288)
Observations[a]	30,500	30,500	30,500	30,500	30,500

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of firms that employ 45-55 workers. The dependent variable is defined as in Equation 9. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

Table 8. Change in Female Share of Employment, Role of Manager's Gender
Employment RD

	All		Manufacturing	
	Female	Male	Female	Male
$FMLA$	-0.676	0.559	-0.028	1.254
	(0.694)	(0.440)	(1.489)	(0.933)
ΔIP_j	0.385	0.485**	0.453	0.629***
	(0.266)	(0.246)	(0.299)	(0.213)
$FMLA \times \Delta IP_j$	-0.482	-0.726**	-0.514	-0.631**
	(0.308)	(0.291)	(0.326)	(0.261)
Observations[a]	13,500	16,500	1,700	3,800

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of firms that employ 45-55 workers separately for firms where at least one of the three top earners is female (Female) and where all three top earners are male (Male), for all firms and manufacturing firms, respectively. The dependent variable is defined as in Equation 9. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

Table 9. Change in Female Share of Employment
Distance RD

	All	24 or less	25-35	36-45	46+
$FMLA$	0.238	0.061	0.838	-0.294	-0.368
	(0.759)	(0.436)	(0.597)	(0.760)	(0.647)
ΔIP_j	2.324**	0.127	-0.865**	1.668**	1.394**
	(0.472)	(0.155)	(0.280)	(0.271)	(0.639)
$FMLA \times \Delta IP_j$	-10.33**	-6.464*	-0.114	-2.557*	-1.198
	(4.032)	(3.546)	(2.004)	(1.530)	(1.503)
Observations[a]	750	750	750	750	750

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of multi-unit firms with two establishments which are 65-85 miles apart. The dependent variable is defined as in Equation 9. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and firm employment, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

Table 10. Change in Female Share of Earnings, By Age
Employment RD

	All	24 or less	25-35	36-45	46+
$FMLA$	-0.197	0.022	-0.111	0.085	-0.194
	(0.442)	(0.168)	(0.338)	(0.267)	(0.287)
ΔIP_j	0.152	-0.028	0.114	0.080	-0.015
	(0.272)	(0.067)	(0.084)	(0.142)	(0.273)
$FMLA \times \Delta IP_j$	-0.188	-0.045	-0.288***	-0.082	0.227
	(0.291)	(0.067)	(0.102)	(0.148)	(0.266)
Observations[a]	30,500	30,500	30,500	30,500	30,500

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of firms that employ 45-55 workers. The dependent variable is defined as in Equation 9. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

Table 11. Change in Female Share of Promotions, By Age
Employment RD

	All	24 or less	25-35	36-45	46+
$FMLA$	-1.833*	-0.863	0.282	-0.158	-1.093
	(1.058)	(0.574)	(0.606)	(0.656)	(0.784)
ΔIP_j	1.438**	-0.372	1.014***	-0.240	1.035
	(0.620)	(0.249)	(0.366)	(0.391)	(0.754)
$FMLA \times \Delta IP_j$	-1.588**	0.429	-1.230***	0.170	-0.958
	(0.751)	(0.316)	(0.422)	(0.452)	(0.797)
Observations[a]	26,500	26,500	26,500	26,500	26,500

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of firms that employ 45-55 workers. The dependent variable is defined as in Equation 9. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

Table 12. Change in Female Share of Employment, By Skill and Age
Employment RD

Panel A: College				
	All	25-35	36-45	46+
$FMLA$	0.257	0.105	0.233	-0.006
	(0.253)	(0.187)	(0.143)	(0.153)
ΔIP_j	-0.031	0.031	-0.015	-0.060
	(0.107)	(0.044)	(0.058)	(0.071)
$FMLA \times \Delta IP_j$	0.070	-0.104	0.043	0.176*
	(0.137)	(0.063)	(0.083)	(0.094)

Panel B: Less than College				
	All	25-35	36-45	46+
$FMLA$	-0.258	-0.459*	0.296	-0.072
	(0.385)	(0.255)	(0.230)	(0.272)
ΔIP_j	0.399*	0.195*	0.304*	0.027
	(0.239)	(0.106)	(0.174)	(0.225)
$FMLA \times \Delta IP_j$	-0.603**	-0.270***	-0.349*	0.021
	(0.267)	(0.100)	(0.194)	(0.246)
Observations[a]	30,500	30,500	30,500	30,500

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of firms that employ 45-55 workers. The dependent variable is defined as in Equation 9. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

Table 13. Change in Female Share of Earnings, By Skill and Age
Employment RD

Panel A: College				
	All	25-35	36-45	46+
$FMLA$	0.461*	0.388	0.047	0.071
	(0.264)	(0.245)	(0.171)	(0.206)
ΔIP_j	0.020	-0.008	-0.079	0.087
	(0.215)	(0.054)	(0.049)	(0.219)
$FMLA \times \Delta IP_j$	0.036	-0.068	0.113	0.033
	(0.243)	(0.069)	(0.078)	(0.237)

Panel B: Less than College				
	All	25-35	36-45	46+
$FMLA$	-0.658*	-0.499**	0.037	-0.265
	(0.388)	(0.228)	(0.236)	(0.246)
ΔIP_j	0.132	0.122**	0.159	-0.102
	(0.220)	(0.062)	(0.127)	(0.188)
$FMLA \times \Delta IP_j$	-0.224	-0.220***	-0.195	0.194
	(0.229)	(0.079)	(0.130)	(0.205)
Observations[a]	30,500	30,500	30,500	30,500

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of firms that employ 45-55 workers. The dependent variable is defined as in Equation 9. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

Table 14. Change in Female Share of Promotions, By Skill and Age
Employment RD

Panel A: College				
	All	25-35	36-45	46+
$FMLA$	-1.320**	-0.468	-0.441	-0.452
	(0.665)	(0.361)	(0.383)	(0.396)
ΔIP_j	0.189	0.184	0.050	-0.048
	(0.431)	(0.250)	(0.215)	(0.165)
$FMLA \times \Delta IP_j$	-0.352	-0.379	-0.081	0.050
	(0.382)	(0.271)	(0.204)	(0.172)

Panel B: Less than College				
	All	25-35	36-45	46+
$FMLA$	-0.512	0.750	0.283	-0.641
	(0.980)	(0.573)	(0.554)	(0.664)
ΔIP_j	1.249*	0.830**	-0.289	1.084
	(0.737)	(0.325)	(0.420)	(0.747)
$FMLA \times \Delta IP_j$	-1.236	-0.850**	0.252	-1.008
	(0.905)	(0.381)	(0.526)	(0.793)
Observations[a]	26,500	26,500	26,500	26,500

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of firms that employ 45-55 workers. The dependent variable is defined as in Equation 9. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.

A Impacts of Import Competition on Output and Relative Female Employment: Numerical Illustration

This section illustrates the effect of an increase in import competition on the relative employment of female workers at FMLA firms ($\lambda = 1$) through its impact on the relative expected cost of employment. We begin by simplifying our set up with the assumption that $\alpha_F = \alpha_M$, so that there is no difference in the productivity of female and male workers. Next, we set $\rho = 1/2$. Demand is linear and downward sloping and the inverse demand curve for the firm is given by

$$P = a - bY. \tag{13}$$

From Equation 3,

$$\frac{L_F^*}{L_M^*} = \left[\frac{W_M(Y)}{W_F(Y)}\right]^2. \tag{14}$$

Substituting for L_F^* in the production function, we obtain

$$Y = L_M^* \left[1 + \frac{W_M(Y)}{W_F(Y)}\right]^2, \tag{15}$$

and therefore

$$L_M^* = Y \left[\frac{W_F(Y)}{W_F(Y) + W_M(Y)}\right]^2, \tag{16}$$

and

$$L_F^* = Y \left[\frac{W_M(Y)}{W_F(Y) + W_M(Y)}\right]^2. \tag{17}$$

Total cost is given by

$$C(Y) = W_M(Y)L_M^* + W_F(Y)L_F^* = Y\frac{W_M(Y)W_F(Y)}{W_M(Y) + W_F(Y)}. \tag{18}$$

The profit maximization problem is thus

$$max_Y Y(a - bY) - C(Y). \tag{19}$$

For parameters such that the second-order conditions are satisfied, the first order condition for profit maximization yields optimal output Y^*. Substituting for optimal output in Equation 3 yields optimal relative female employment as a function of the parameters a, b, g, τ, γ_F, γ_M, w_F, w_M and parameters of $\theta(Y)$.

We posit that an increase in import competition is associated with a downward shift of the demand curve such that a decreases. Figures A.1 and A.2 provide a numerical example to illustrate the impact of import competition. We trace the optimal output and ratio of female to male workers as a function of the demand parameter a. We set $b = 1/100$, $\gamma_F = 0.2$ and $\gamma_M = 0.1$. The rationale behind choosing γ_F to be greater than γ_M is that firms may

perceive female workers to place a higher value on leave, and associate a greater retention benefit with them. We set $w_F = 0.8$ and $w_M = 1$. Hence, while the male wage is normalized to one, the female wage is about 80 percent of the male wage, consistent with the gender wage gap in the U.S. in the early 2000s (Blau and Kahn, 2017). Further, we equate $g = 0.1$ and set the gender differential in the firm's belief about leave-taking to $\tau = 2$, in line with evidence discussed in Section 2 which suggests that women with young children are almost 1.7 times more likely to take leave under the FMLA compared to their male counterparts. Finally, we define $\theta(Y) = \frac{1}{1+Y^1/3}$. The figures show a positive relationship between a and both output and the relative employment of female workers. The decrease in demand from greater import competition is therefore associated with lower relative employment of female workers.

Figure A.1. Import Competition and Output

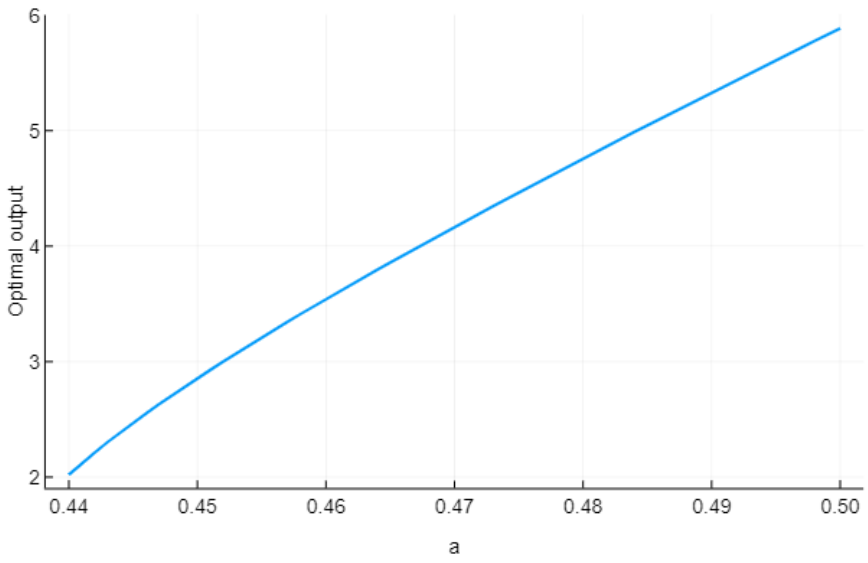


Figure A.2. Import Competition and Relative Employment of Female Workers

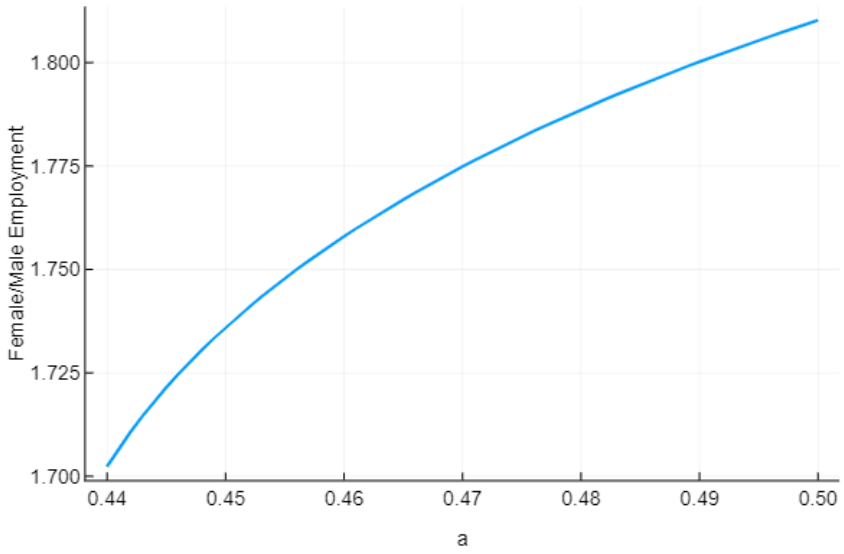
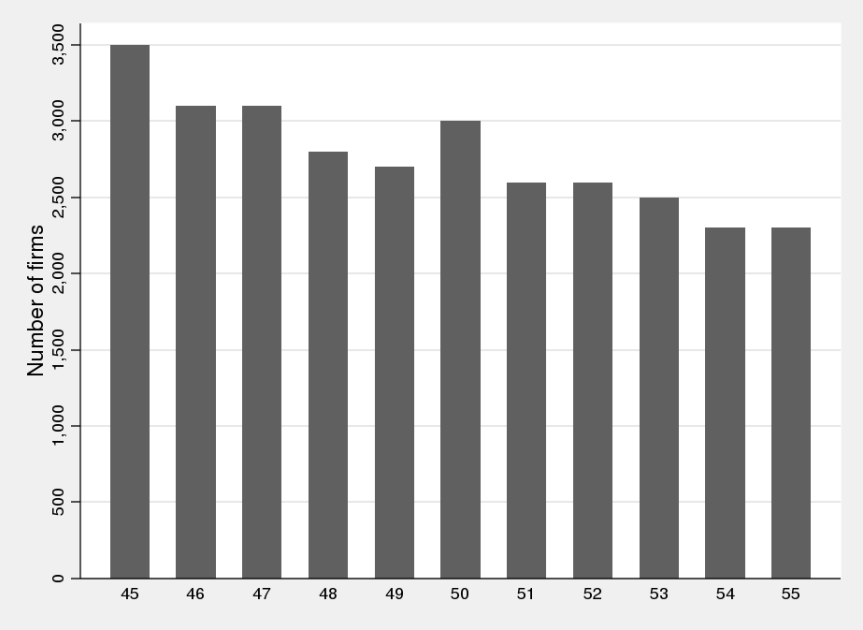


Figure A.3. Employment Bunching



Notes: Bars indicate number of firms in a given employment bin between 45 and 55 as of 2000. Number of firms rounded to comply with Census Bureau disclosure avoidance rules.

Source: Author's calculations using LBD.

Table A-1. Change in Female Share of Employment by Skill
Distance RD

	College	Less than College
$FMLA$	-1.053*	1.290*
	(0.561)	(0.773)
ΔIP_j	0.523*	1.802**
	(0.305)	(0.283)
$FMLA \times \Delta IP_j$	-2.169	-8.164*
	(1.377)	(4.303)
Observations[a]	750	750

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered by four-digit NAICS. This table displays results from estimating Equation 8 on the sample of multi-unit firms with two establishments which are 65-85 miles apart. The dependent variable is defined as in Equation 9. All regressions include a quadratic control function, $f(r_i)$, in the rating variable, pre-shock controls for firm age and multi-unit status, share of female workers in a four-digit NAICS, and state fixed effects.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using LEHD and LBD.